Bindita Chaudhuri

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Summary

Senior Applied Research Scientist with 10+ years of expertise (5+ years industrial R&D) steering the development and commercial product deployment of multimodal Generative AI systems. Expertise includes post-training audio-visual foundation models for speech-driven facial animation, neural avatars using GANs and NeRFs, scene text editing using DiTs, audio refinement, controllable motion generation and photorealistic reconstruction of 3D humans.

Proven track record of leading technical roadmaps and mentoring junior researchers at Meta Reality Labs, Microsoft, and Flawless AI, backed by Ph.D. in Computer Vision and primary-author publications at top-tier venues (CVPR, ECCV etc.).

Technical Skills

- **Generative AI**: Diffusion Transformers (data curation & distributed training), GANs, VAEs, 3D Gaussian Splatting.
- **Deep Learning**: PyTorch, TensorFlow, Keras.
- **Core Languages & Tools**: Python, C/C++, CUDA, Swift.
- **Domains**: Multimodal (Audio-Visual) Foundation Models, 3D Computer Vision, Digital Humans, Facial Animation.

Work Experience

Flawless AI, *Senior Applied Scientist/Applied Science Manager* — 2023 - Present, Los Angeles, CA

- Architected a multilingual audio-driven expressive 3D facial animation pipeline by distributed supervised finetuning of large audio-visual foundation models, including targeted synthetic data generation for data curation and hybrid quantitative-perceptual evaluation protocols.
- Redesigned the neural rendering GAN architecture to improve photorealism and temporal consistency for Hollywood-grade filmmaking and VFX (used in ***Watch the Skies* on Amazon Prime Video**). Hardened the pipeline against edge cases (occlusions, film grain, extreme poses), improving robustness in real-world production environments by 23%.
- Engineered an end-to-end video scene text editing pipeline using OCR, SAM2 and DiTs, establishing a new product capability for automated localization.
- Improved 3D face modeling and tracking for beards using high-dimensional Gaussian Splatting.
- Deployed speaker diarization and crosstalk separation for audio refinement capabilities.
- Supervised an intern project that introduced text-controlled 3D human body motion generation and editing frameworks, lowering manual animation time for artists (*ICCV 2025*).
- Led a team of 5+ scientists and coordinated priorities across engineering and product teams, defining the technical roadmaps from foundational research to production deployment.

Meta Reality Labs, *Research Scientist* — 2021 – 2023, Redmond, WA

- Scaled and optimized Masked Generative Transformers for text/image-to-video synthesis using FSDP and mixed-precision training strategies, improving throughput by 11%.
- Designed TalkinNeRF, an animatable NeRF for reconstructing full-body talking humans with fine-grained hand and face details from text and speech inputs (*ECCVW 2024*).
- Mentored an intern on a novel multimodal coarse-to-fine approach for detailed 4D face geometry reconstruction from in-the-wild videos for AR/VR applications (*CVPR 2023*).
- Architected a spatio-temporal deep network to accumulate information from dynamic in-the-wild videos, achieving self-occlusion-free face texture completion.
- Advanced speech-driven 3D facial animation by integrating neural processing of emotion and non-speech vocalizations, drastically improving realism in 3D video calling.
- Spearheaded enhanced texture recovery for synthetic faces from challenging videos utilizing a highly customized StyleGAN-based architecture.

Facebook Reality Labs Research, *Research Intern* — Summer 2020, Sausalito, CA

- Designed a region-adaptive VAE to synthesize photorealistic texture maps for 3D humans with independent geometry and appearance controllability targeting virtual try-on applications and semi-supervised synthetic data generation (*CVPR 2021*).

Microsoft Cloud & AI, *Student Researcher* — 2019 - 2020, Redmond, WA

- Developed a high-fidelity personalized 3D face avatar generation pipeline from user videos, reducing identity drift (*ECCV 2020 Spotlight*).
- Stabilized real-time face tracking algorithms optimized for challenging, unconstrained in-the-wild video streams.

Microsoft Research, *Research Intern* — Summer 2018, Redmond, WA

- Designed a real-time multi-task learning framework for joint 2D face detection and 3D facial motion retargeting (*CVPR 2019*); successfully shipped the model to production for **SwiftKey Puppets**, scaling to millions of active Android users for immersive virtual communication.

Intel Labs, *Research Intern* — Summer 2017, Santa Clara, CA

- Implemented a Spatial Transformer Network for optical flow prediction and image super-resolution to enable high-quality frame interpolation from HD multi-camera arrays. U.S. patent granted.

Education

Ph.D. — **University of Washington**, *Computer Science* — 2016 – 2021
- *Thesis*: Deep Facial Expression Modeling and 2D-to-3D Motion Retargeting

M.E. — **Indian Institute of Technology Bombay**, *Electrical Engineering* — 2014 – 2016
- *Honors*: Department Academic Excellence Award
- *Thesis*: Region-based Remote Sensing Image Retrieval using Graph Theory

B.E. — **Jadavpur University**, *Electronics and Telecommunication Engineering* — 2010 – 2014
- *Honors*: University Gold Medal, The Supriya Basu Scholarship & others
- *Thesis*: Low Cost Low Bandwidth Virtual Education Platform Design

Selected Publications

KinMo: Kinematic-aware Human Motion Understanding and Generation [link]	ICCV 2025
TalkinNeRF: Animatable Neural Fields for Full-Body Talking Humans [link]	ECCVW 2024
Semi-supervised Synthesis of High-Resolution Editable Textures for 3D Humans [link]	CVPR 2021
Personalized Face Modeling for Improved Face Reconstruction and Motion Retargeting [link]	ECCV 2020 (Spotlight)
Joint Face Detection and Facial Motion Retargeting for Multiple Faces [link]	CVPR 2019
Learning to Generate 3D Stylized Character Expressions from Humans [link]	WACV 2018
Deep Learning for Multi-Camera View Interpolation [link]	U.S. Patent *10,547,823*, 2020

Professional Activities

- **Conference Reviewer**: CVPR, ECCV, ICCV, NIPS, SIGGRAPH Asia, BMVC, ICLR, ICPR, IEEE VR
- **Journal Reviewer**: ACM TOG, IEEE TPAMI, CVIU, IEEE TGRS, IEEE GRSL
- **Invited Talk**: Beyond PhD Conference, University of Southern California
- **Area Chair (Student)**: CSE Graduate Admissions Committee, University of Washington (2020).
- **Teaching Assistant**: University of Washington (CSE) and IIT Bombay (EE).
- People's Choice Award (University of Washington Research Day, 2017)